

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Milestone 3 scribe

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Question-1: System Objective and Deterministic Baseline

Solution:

The objective of this project is to move beyond single “best guess” estimates of glacier melting by quantifying the uncertainty propagation within a physical model.

- **The System:** A model of the processes of meteorological and physical variables to determine glacier mass balance.
- **The Problem:** While physical models can be well-calibrated, there is significant random and systematic uncertainty in climatic inputs like temperature and albedo.
- **The Solution:** We use Monte Carlo simulations to translate this climate variability into a Probability Density Function (PDF) of the melt rate.

Question-2: Random Variables in the Simulation**Solution:**

We model five primary environmental variables to capture climatic uncertainty. Each is assigned a distribution that matches its physical behavior.

0.1 Air Temperature (T_a)

- Distribution: Normal distribution

$$T_a \sim \mathcal{N}(\mu, \sigma^2)$$

- Parameters: Mean $\mu = 0$ and $\sigma \approx 0.424^\circ C$
- Modeling: Uncertainty is modeled by adding both systematic (ϵ_s) and random (ϵ_r) error terms to the measured temperature.

0.2 Ice Albedo (α_i)

- Distribution: Truncated Normal Distribution
- Bounded Range: Strictly between 0.2 and 0.6
- Parameters: Mean $\mu = 0.34$, $\sigma = 0.05$

This prevents the simulation from selecting impossible values outside the physical range.

0.3 Precipitation (P)

- Distribution: Gamma Distribution

$$P \sim \text{Gamma}(k, \theta)$$

- Parameters: Shape $k = 2.5$, Scale $\theta = 3.0$
- Logic: The Gamma distribution ensures precipitation values remain non-negative ($P \geq 0$).

0.4 Cloudiness (n)

- Distribution: Normal Distribution

$$n \sim \mathcal{N}(0.5, 0.2^2)$$

- Logic: Values represent sky coverage and are clipped between 0 and 1.

0.5 Snowfall Threshold (T_s)

- Distribution: Normal Distribution

$$T_s \sim \mathcal{N}(1.0, 0.5^2)$$

- Role: Acts as a transition variable that determines if precipitation falls as snow or rain.

Question-3: Probabilistic Modeling and Fixed Decision Rule**Solution:**

The “fixed rule” of our system is the Surface Energy Balance Model. Even though the inputs are random, the calculation method remains deterministic.

Step A: Surface Melt Energy (Q_m)

The variables are inserted into the energy equation to find the energy available for melt:

$$Q_m = (1 - \alpha)S_{in} + C_0 + C_1T_a$$

- α : Samples from the truncated albedo distribution
- T_a : Samples from the air temperature distribution
- S_{in} : Incoming solar radiation
- C_0, C_1 : Fixed thermodynamic constants

Step B: Mass Balance Conversion (M)

The melt energy is converted into melt mass (or melt depth):

$$M = \left(\frac{Q_m}{\rho_w L_f} \right) \Delta t$$

- ρ_w : Density of water
- L_f : Latent heat of fusion
- Δt : Time step of the simulation

Question-4: Simulation Pipeline**Solution:**

The system operates as a multi-stage uncertainty propagation pipeline.

- (a) **Input Generation:** Generate $N \geq 10,000$ samples for each random variable using their PDFs.
- (b) **Stochastic Perturbation:** Add systematic uncertainty (ϵ_s) and random uncertainty (ϵ_r).
- (c) **Physical Processing:** Inputs pass through the deterministic equations for Q_m and M .
- (d) **Output Analysis:** Thousands of simulation results are used to construct final PDF and CDF outputs. The Weak Law of Large Numbers ensures convergence to the true mean.

Question-5: Simulation Outputs and Interpretation**Solution:****Performance Metrics:**

The deterministic baseline simulation is evaluated using several statistical performance metrics:

- **Mean and Variance:** These are used to determine the central tendency and the overall spread of the seasonal glacier melt estimates.
- **Concentration Bounds:** Chebyshev's inequality is applied to establish distribution-free probabilistic limits (e.g., $P(|X - \mu| \geq k\sigma) \leq 1/k^2$), providing a rigorous boundary for mass balance risk.
- **Convergence Stability:** The Weak Law of Large Numbers (WLLN) is used to ensure that the simulation results stabilize as the number of Monte Carlo iterations ($N \geq 10,000$) increases.

PDF/CDF Plots and Statistical Summaries:

The simulation generates Probability Density Functions (PDF) and Cumulative Distribution Functions (CDF) to represent the variability of glacier melt. For instance, a verification histogram for Ice Albedo (bounded between 0.2 and 0.5 in initial tests) confirms that the Monte Carlo sampler correctly adheres to physical constraints.

Observed Behaviors:

Under the deterministic strategy, the system exhibits high sensitivity to variations in Air Temperature (T_a) and Ice Albedo (α_i). The propagation of these uncertainties through the energy balance equations results in a wide distribution of possible melt outcomes, highlighting that extreme melting events are often driven by the co-occurrence of low albedo and high temperatures.

Question-6: Technical Gaps and Preparation for Next Stage**Solution:****Current Limitations of the Deterministic Baseline:**

The primary limitation of the current baseline is its reliance on a single-point spatial representation, which does not account for the varying topography and physical conditions across an entire glacier surface.

Simplified Aspects:

Several key physical processes are currently simplified:

- **Independence Assumption:** Climatic variables (e.g., temperature and cloudiness) are treated as independent random variables, neglecting real-world meteorological correlations.
- **Stationarity:** Uncertainty is assumed to be stationary over the simulation horizon, ignoring potential long-term climate shifts.
- **Linearization:** The energy balance formulation is largely linearized to simplify the conversion of energy to melt mass.

Role of Randomness in Future Milestones:

In upcoming stages, introducing correlated randomness will allow the model to capture complex dependencies, such as the relationship between cloud cover and radiative transfer. This will improve the reliability of the probabilistic estimates by reflecting more realistic atmospheric conditions.

Planned Refinements for Milestone 4:

For Milestone 4, we plan to transition from point-scale to spatial glacier modeling and implement Bayesian estimation to refine our distribution parameters based on actual observational data. Additionally, we will incorporate non-linear albedo feedback loops to model how melting ice further darkens the surface, accelerating future melt rates.

End of Submission